

This analysis examines whether transcriptomic differences across organ-tropic metastatic sublines reflect adaptation to distinct microenvironments. Results are presented for dimensionality reduction and differential expression analyses.

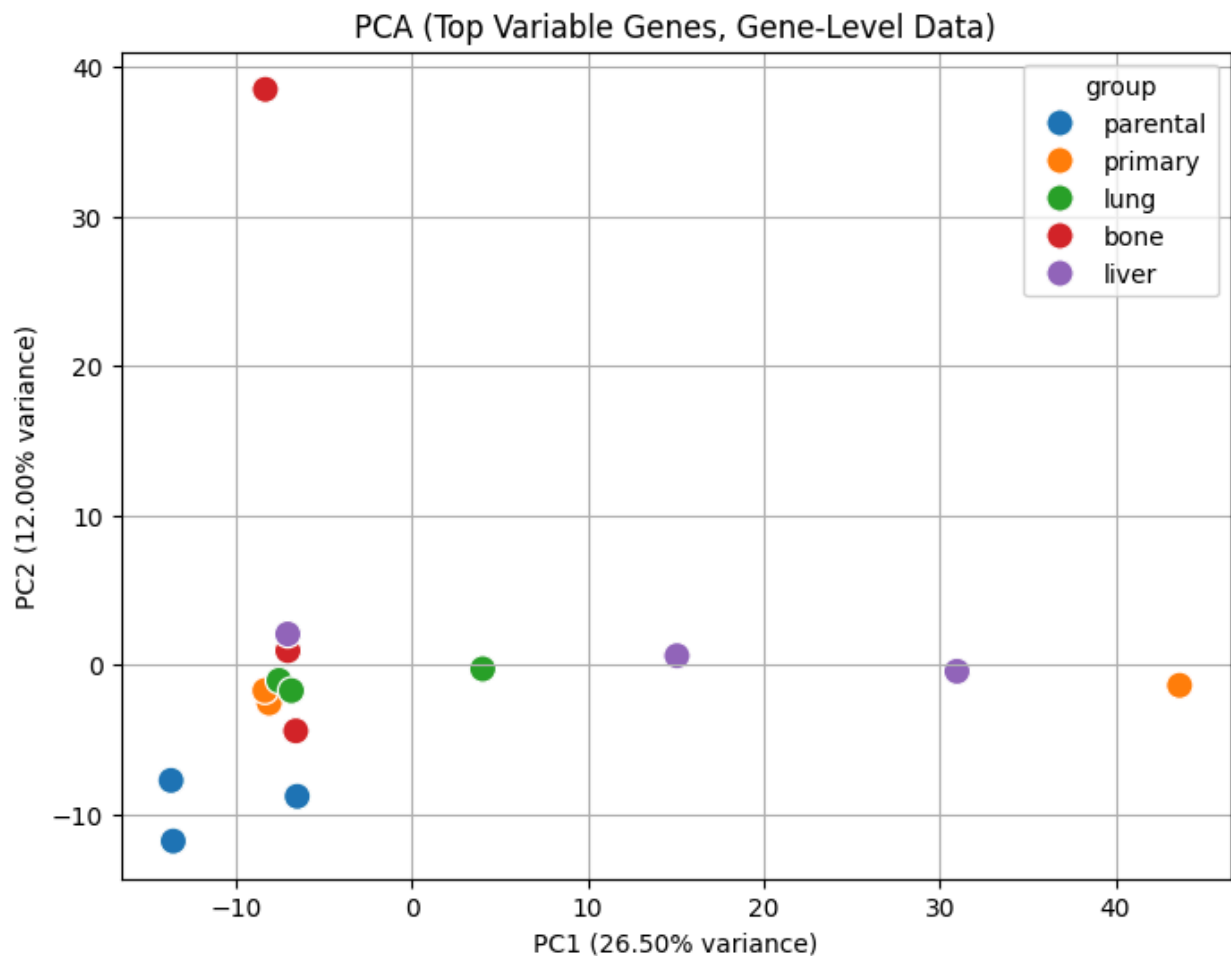


Figure 1: PCA of gene-level expression (top 1000 most variable genes) showing distribution of metastatic sublines. PC1 (26.5% variance) captures an axis along which liver-tropic samples are spread rather than tightly clustered. Lung samples cluster closely near the origin. Bone samples show pronounced heterogeneity, including one clear outlier along PC2 (~38).

Primary tumor samples also show variability, with one sample well separated along PC1. With $n=3$ per group, these patterns are descriptive rather than conclusive — they are consistent with more variable transcriptional states among bone-tropic samples than liver-tropic ones, but a larger cohort would be needed to establish this as a real biological difference rather than sampling noise.

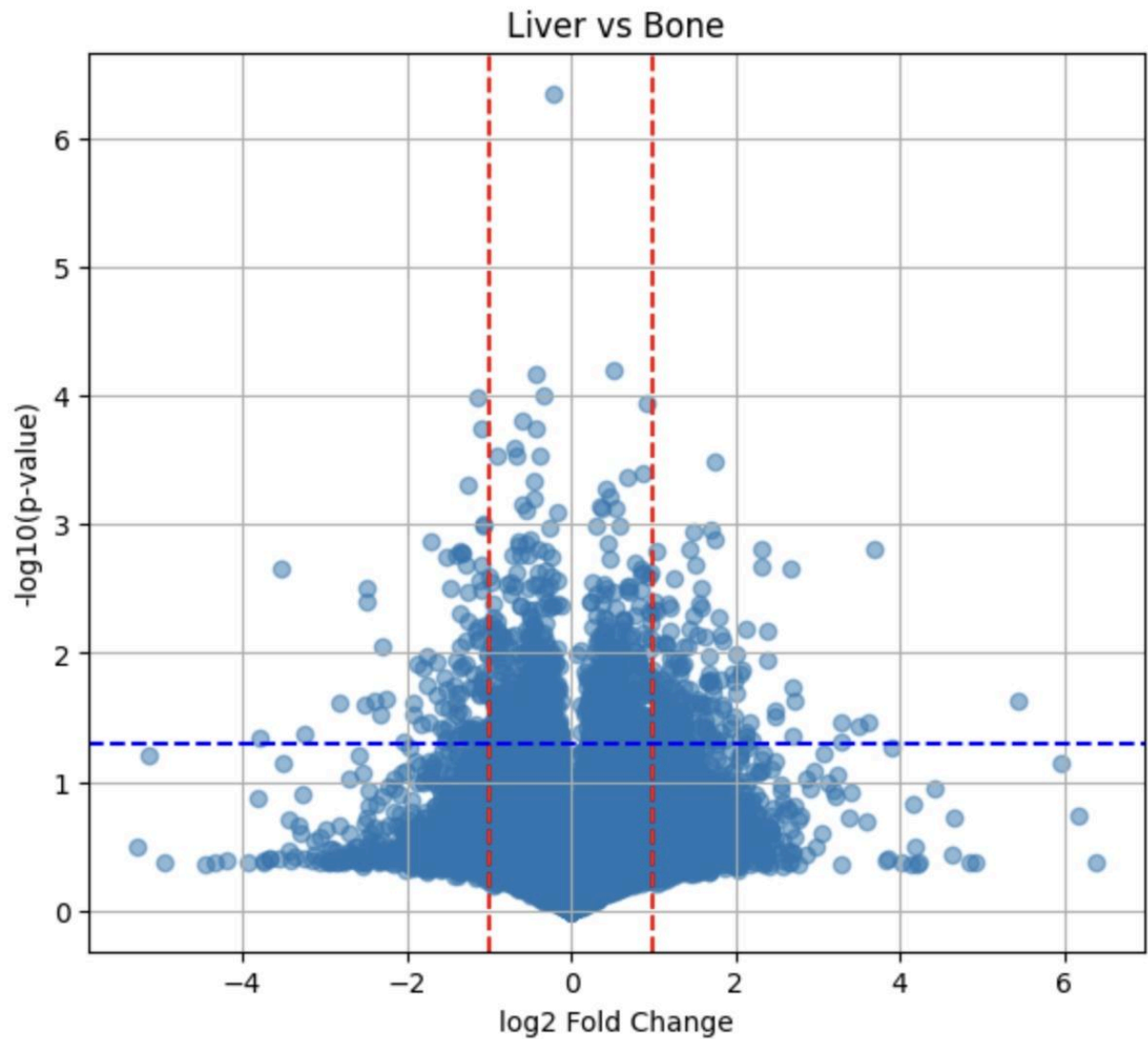


Figure 2: Volcano plot comparing liver- and bone-tropic metastatic sublines. Log2 fold changes span a broad range (approximately -5 to +6), and more genes exceed the fold-change thresholds here than in the liver vs lung comparison. No genes met FDR significance at this sample size. The wider spread of fold changes is a descriptive observation about this comparison, not evidence of a specific biological mechanism — it's a pattern worth following up in a properly powered dataset, not a finding in itself.

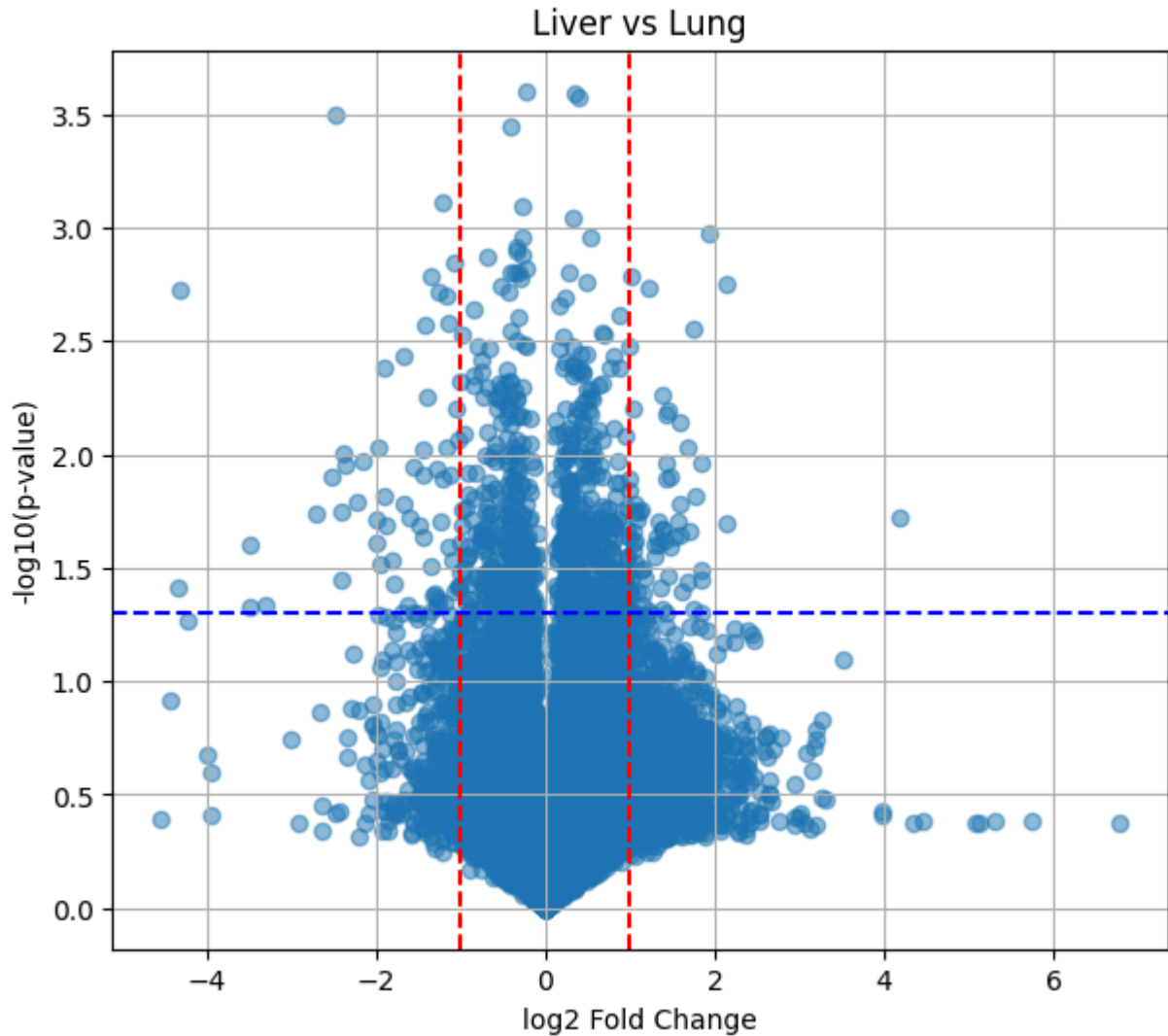


Figure 3: Volcano plot comparing liver- and lung-tropic metastatic sublines. Most genes cluster near $\log_2$ fold change ≈ 0 , and fewer genes exceed fold-change thresholds compared to the liver vs bone comparison. This is consistent with a smaller transcriptional difference between liver and lung sublines than between liver and bone, though as with Figure 2, no genes reached significance after correction, so this remains an exploratory observation rather than a confirmed result.

Together, these results are exploratory: with $n=3$ per group and no genes reaching significance after multiple-testing correction, they cannot establish a mechanism of organ-specific metastatic adaptation. What they do provide is a specific, testable pattern — smaller transcriptional distance between liver and lung sublines than between liver and bone — worth investigating in a larger, adequately powered dataset.